Data annotation attributes provide a way for you to add additional information, or *metadata*, to types in .NET, including specific rules the data should honor, such as business requirements related to presence, data type, and range. The validation framework is one area among many in the .NET Framework that is responsive to the hints provided by data annotations. You will explore how to take advantage of this capability to further eliminate boilerplate code when building your forms and easily validate your data against your business rules.

The examples in this chapter build on the CityBreaks application that was started in the last chapter. We'll assume you have configured the application to use the lowercase options for URLs and implemented both parameter transformers, so routes and parameters use kebab casing. The downloads for this chapter include a version that has these settings applied if you require a starting point.

5.1 Forms basics

Forms consist of an HTML `form` tag that encloses a number of controls for gathering user input and, usually, a control of some description that enables the user to submit the content of the input controls to the server for processing. The form submission itself results in an HTTP request. The request will use whichever HTTP method is specified by the form element's `method` attribute, with `GET` being the default if no method is specified. When the `GET` method is used, the form values are appended to the URL as key-value pairs in a query string (see figure 5.1). The key is generated from the `name` attribute of the form control, and the value is obtained from the control. When the `POST` method is used, the content of the form is included in the body of the request as key-value pairs constructed using the same pattern as the `GET` method.